Code

Main Class

```
package Phase1Final;
import java.util.*;
public class RentalCamera_MainFile {
	static Scanner sc = new Scanner(System.in);
	static List<Camera> cameraList = new ArrayList<>();
	static List<User> users = new ArrayList<>();
	static Map<String, Double> wallet = new HashMap<>();
	static String loggedInUser = null;
	public static void main(String[] args) {
		cameraList.add(new Camera(1, "Samsun3g", "DS123",500.0, "Available"));
		cameraList.add(new Camera(2, "Sony", "HD214", 500.0, "Available"));
		cameraList.add(new Camera(3, "Panasonic", "XC", 500.0,"Available"));
		cameraList.add(new Camera(4, "Canon", "XLR", 500.0,"Available"));
		cameraList.add(new Camera(5, "Fujitsu", "J5", 500.0, "Available"));
		cameraList.add(new Camera(7, "Sony", "HD226", 500.0, "Available"));
		cameraList.add(new Camera(8, "Samsung", "DS246",500.0, "Available"));
		cameraList.add(new Camera(9, "LG", "L123", 500.0, "Available"));
		cameraList.add(new Camera(10, "Canon", "XPL",500.0, "Available"));
		cameraList.add(new Camera(11, "Chroma", "CT", 500.0, "Available"));
		cameraList.add(new Camera(12, "Canon", "Digital", 123.0, "Available"));
		cameraList.add(new Camera(13, "NIKON", "DSLR-D7500", 500.0, "Available"));
		cameraList.add(new Camera(14, "Sony", "DSLR12", 200.0, "Available"));
		cameraList.add(new Camera(15, "Samsung", "SM123", 200.0, "Available"));
		cameraList.add(new Camera(16, "Sony", "SONY1234", 123.0, "Available"));
```

```
users.add(new User("naresh", "rao", 10000));
System.out.println("+-----------------------------------------+");
System.out.println("| WELCOME TO CAMERA RENTAL |");
System.out.println("+-----------------------------------------+");
System.out.println("PLEASE LOGIN TO CONTINUE -");
// login
while (true) {
try {

System.out.print("USERNAME - ");
String username = sc.nextLine();

System.out.print("PASSWORD - ");
String password = sc.nextLine();
for (int i = 0; i < users.size(); i++) {
User user = users.get(i);

if(user.getUsername().equalsIgnoreCase(username) && user.getPassword().equals(password)) {

loggedInUser = username;
break;
}
}
```

```
                    if (loggedInUser ==
null) {

        System.out.println("Invalid credentials. Please try again.");
                    } else {
                        break;
                    }
                } catch (Exception e) {

        System.out.println("Please Enter creds in alphanumeric & valid ones\n");
                }
            }
            // main menu
            while (true) {
                System.out.println("1. MY
CAMERA");
                System.out.println("2. RENT
A CAMERA");
                System.out.println("3. VIEW
ALL CAMERAS");
                System.out.println("4. MY
WALLET");
                System.out.println("5. EXIT");
                try {
                    int choice =
Integer.parseInt(sc.nextLine());
                    switch (choice) {
                    case 1:

        myCamera(loggedInUser);
```

```
                    break;
                case 2:
viewAllCameras();
rentCamera(loggedInUser);
                    break;
                case 3:
viewAllCameras();
                    break;
                case 4:
myWallet(loggedInUser);
                    break;
                case 5:
System.out.println("Thank you for using the app!");
System.exit(0);
                    default:
System.out.println("Invalid choice. Please try again.");
break;
                }
            } catch
(NumberFormatException | InputMismatchException
                    e) {
```

```
System.out.println("Please enter valid number from 1 to 5\n");
                                                                    }
                                                                }
}
public static void myWallet(String loggedInUser) {
        for (int i = 0; i < users.size(); i++) {
                User user = users.get(i);
                if (user.getUsername().equalsIgnoreCase(loggedInUser)) {
                        System.out.println("YOUR CURRENT WALLET BALANCE IS - INR." +
user.getWallet());
                                                System.out.print("DO YOU WANT TO DEPOSIT
MORE AMOUNT TO YOUR WALLET?(1.YES 2.NO) - ");
                                                                        String input =
sc.nextLine();
                                                                if (input.equals("1")) {
                                                                    try {

System.out.print("ENTER THE AMOUNT (INR) - ");
                                                                                    String
input2 = sc.nextLine();
                                                                                    int
amountUpdate = Integer.parseInt(input2);
                                                                                    // Find
the user with the specified username
                                                                                    User
userToUpdateWallet = null;
                                                                                    for (int
j = 0; j < users.size(); j++)
                                                                                    {
```

```
User user1 = users.get(j);

if

(user.getUsername().equalsIgnoreCase(loggedInUser)) {

userToUpdateWallet = user1;

break;

}

}

//
Update the user wallet with money

if
(userToUpdateWallet != null) {

userToUpdateWallet.addMoneyToWallet(amountUpdate);

System.out.println("YOUR WALLET BALANCE UPDATED SUCCESSFULLY. CURRENT WALLET
BALANCE - INR."+ user.getWallet());

} else {

System.out.println("User with username " + loggedInUser + " not found.");

}

} catch
(NumberFormatException |

InputMismatchException e) {

System.out.println("\nPlease enter valid amount in numbers");
```

```
                                                        }
                                                    } else {
                                                        break;
                                                    }
                    }
            }
    }
    public static void myCamera(String loggedInUser) {
            Scanner sc = new Scanner(System.in);
            boolean backToMenu = false;
            while (!backToMenu) {
                    try {
                            System.out.println("1. ADD");
                            System.out.println("2. REMOVE");
                            System.out.println("3. VIEW MY CAMERAS");
                            System.out.println("4. GO TO PREVIOUS MENU");
                            // System.out.print("Enter your choice: ");
                            int choice = sc.nextInt();
                            switch (choice) {
                            case 1:
                                    addCamera();
                                    break;
                            case 2:
                                    removeCamera();
                                    break;
                            case 3:
                                    viewMyCameras(loggedInUser);
```

```
                    break;
                case 4:
                    backToMenu = true;
                    break;
                default:
                    System.out.println("Invalid choice. Please try again.");
                        break;
                }
            } catch (InputMismatchException e) {
                System.out.println("\nPlease enter valid number from 1 to 4\n");
                    break;
            }
        }
        // sc.close();
    }
    @SuppressWarnings("resource")
    private static void rentCamera(String loggedInUser) {
        Scanner sc = new Scanner(System.in);
        try {
            // Get user input
            System.out.print("ENTER THE CAMERA ID YOU WANT TO RENT -");
                    int cameraCode = sc.nextInt();
                    // Get the camera from the camera list
                    Camera getcamera = getCameraById(cameraList,
cameraCode);
                    if (getcamera == null) {
                        System.out.println("Camera with ID " +
cameraCode +
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                                                    " not found.");
                                        return;
                                }
                                // Get rental period
                                System.out.print("ENTER RENTAL PERIOD (in days) - ");
                                int rentalPeriod = sc.nextInt();
                                sc.nextLine(); // Consume the newline character left by
nextInt()
                                getcamera.setStatus("RENTED");
                                double rentAmount = getcamera.getPrice() * rentalPeriod;
                                // Find the user with the specified username
                                User userToUpdate = null;
                                for (int i = 0; i < users.size(); i++) {
                                        User user = users.get(i);
                                        if

        (user.getUsername().equalsIgnoreCase(loggedInUser)) {
                                                userToUpdate = user;
                                                break;
                                        }
                                }
                                if (rentAmount > userToUpdate.getWallet()) {
                                        System.out.println("ERROR : TRANSACTION FAILED
DUE TO INSUFFIECIENT WALLET BALANCE. PLEASE DEPOSIT THE AMOUNT TO YOUR WALLET.");
                                                return;
                                }
                                // Update the myCameras ArrayList of the found user
object
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```
                    if (userToUpdate != null) {
                        userToUpdate.addCamera(getcamera);

        userToUpdate.deductMoneyFromWallet(getcamera.getPrice());
                    }
                    // removing from list after user renting
                    for (Camera camera : cameraList) {
                        if (camera.getId() == cameraCode) {
                            cameraList.remove(camera);
                            break;
                        }
                    }
                    System.out.println("Camera rented successfully for " +
                    rentalPeriod + " days. Your wallet balance is now $"+
                    userToUpdate.getWallet());
                        System.out.println("YOUR TRANSACTION FOR
CAMERA - " +
                                "with rent INR." + rentAmount + "
HAS SUCCESSFULLY COMPLETED.");
            } catch (NumberFormatException | InputMismatchException e) {
                System.out.println("Please enter valid number\n");
            }
            // sc.close();
        }
        public static Camera getCameraById(List<Camera> cameraList, int id) {
            for (Camera camera : cameraList) {
                if (camera.getId() == id) {
                    return camera;
```

```
                }
            }
            return null; // Camera with given id not found in the list
        }
        public static void addCamera() {
            try {
                // System.out.println("\nADD CAMERA");
                System.out.print("ENTER THE CAMERA BRAND - ");
                String brand = sc.nextLine();
                System.out.print("ENTER THE MODEL - ");
                String model = sc.nextLine();
                System.out.print("ENTER THE PER DAY PRICE (INR) - ");
                double price = Double.parseDouble(sc.nextLine());
                int id = generateCameraId();
                Camera camera = new Camera(id, brand, model, price,
                        "Available");
                cameraList.add(camera);
                System.out.println("YOUR CAMERA HAS BEEN SUCCESSFULLY ADDED TO
THE LIST.\n");
            } catch (NumberFormatException | InputMismatchException e) {
                System.out.println("Please enter valid the valid data\n");
            }
        }
        public static int generateCameraId() {
            int lastCameraId = 0;
            if (!cameraList.isEmpty()) {
                Camera lastCamera = cameraList.get(cameraList.size() -
```

```
                                        1);
                        lastCameraId = lastCamera.getId();
                }
                int newCameraId = lastCameraId + 1;
                return newCameraId;
        }
        public static void removeCamera() {
                Scanner scanner = new Scanner(System.in);
                try {



        System.out.println("==========================================================
=========");
                                        System.out.println("CAMERA ID BRAND MODEL PRICE (PER
DAY) STATUS");



        System.out.println("==========================================================
======== ");
                                                for (int i = 0; i < cameraList.size(); i++) {
                                                        Camera camera = cameraList.get(i);
                                                        System.out.format("%-10d%-10s%-
10s%-18s%s%n",
                                                                camera.getId(),
camera.getBrand(), camera.getModel(),
                                                                camera.getPrice(),

        camera.getStatus().equalsIgnoreCase("Available")
                                                                ? "Available" : "Not
Available");
```

```
                                                                    }
                                                                    System.out.println("--------------------------------
----------------------");
                                                                                System.out.print("ENTER THE
CAMERA ID TO REMOVE - ");
                                                                                // Use hasNextInt() method
to check if there is an integer input
                                                                                if (scanner.hasNextInt()) {
                                                                                        int cameraId =
scanner.nextInt();
                                                                                        // process the input
                                                                                        boolean found =
false;
                                                                                        for (int i = 0; i <
cameraList.size(); i++) {
                                                                                                Camera
camera = cameraList.get(i);
                                                                                                if
(camera.getId() == cameraId) {

        cameraList.remove(camera);
                                                                                                        found
= true;

        System.out.println("CAMERA SUCCESSFULLY REMOVED FROM THE LIST - ");

        break;
                                                                                                }
                                                                                        }
                                                                                        if (!found) {
```

```
System.out.println("Camera with ID " + cameraId + " not found.");
                                                                                    }
                                                                        } else {

System.out.println("Invalid input. Please enter an integer.");
                                                                        }
        } catch (NumberFormatException | InputMismatchException e) {
                System.out.println("Please enter valid data\n");
        } finally {
                // scanner.close();
        }
}
public static void viewAllCameras() {
        System.out.println("\nFOLLOWING IS THE LIST OF AVAILABLE CAMERA(S)-");

System.out.println("=========================================================
==========================");
                                        if (cameraList.size() == 0) {
                                                System.out.println("No cameras available
for rent.");
                                        } else {
                                                System.out.println("CAMERA
ID\tBRAND\t\tMODEL\t\tPRICE (PER DAY)\tSTATUS");

System.out.println("=========================================================
==========================");
                                                                                for (Camera
camera : cameraList) {
```

```
System.out.println(camera.getId() + "\t\t" +

camera.getBrand() + "\t\t" + camera.getModel() + "\t\t"

+ camera.getPrice() + "\t\t" +

camera.getStatus());
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
}
}

System.out.println("==========================================================
==========================");
}
public static void viewMyCameras(String loggedInUser) {
for (User user : users) {
if (user.getUsername().equals(loggedInUser)) {
user.viewMyCameras();
return;
}
}
System.out.println("User with username " + loggedInUser + " not found.");
}
}
```

Camera Class

```
package Phase1Final;
public class Camera {
	private int id;
	private String brand;
	private String model;
	private double price;
	private String status;
	public Camera(int id, String brand, String model, double price,
			String status) {
		this.id = id;
		this.brand = brand;
		this.model = model;
		this.price = price;
		this.status = status;
	}
	public int getId() {
		return id;
	}
	public String getBrand() {
		return brand;
	}
	public String getModel() {
```

```
        return model;
    }
    public double getPrice() {
        return price;
    }
    public String getStatus() {
        return status;
    }
    public void setStatus(String status) {
        this.status = status;
    }
    @Override
    public String toString() {
        return String.format("%-10s%-15s%-15s%-20.2f%-15s", id,
                    brand, model, price, status);
    }
}
```

User Class

```
package Phase1Final;
import java.util.ArrayList;

public class User {
    private String username;
    private String password;
    private ArrayList<Camera> myCameras;
    private double wallet;

    public User(String username, String password, double wallet) {
        this.username = username;
        this.password = password;
        myCameras = new ArrayList<>();
        this.wallet = wallet;
    }

    public User(String username, String password, ArrayList<Camera> myCameras,
double wallet) {
        super();
        this.username = username;
        this.password = password;
        myCameras = new ArrayList<>();
```

```
    wallet = 0;
}

public String getUsername() {
    return username;
}

public String getPassword() {
    return password;
}

public ArrayList<Camera> getMyCameras() {
    return myCameras;
}

public void addCamera(Camera camera) {
    myCameras.add(camera);
}

public void setMyCameras(ArrayList<Camera> myCameras) {
    this.myCameras = myCameras;
}

public void removeCamera(int id) {
```

```
        for (Camera camera : myCameras) {
            if (camera.getId() == id) {
                myCameras.remove(camera);
                System.out.println("Camera with ID " + id + " removed from your list.");
                return;
            }
        }
        System.out.println("Camera with ID " + id + " not found in your list.");
    }

    public void viewMyCameras() {
        if (myCameras.isEmpty()) {
            System.out.println("You have no cameras in your list.");
            return;
        }
        System.out.println(String.format("%-10s%-15s%-15s%-20s%-15s", "CAMERA
ID", "BRAND", "MODEL", "PRICE (PER DAY)", "STATUS"));
        for (Camera camera : myCameras) {
            System.out.println(camera);
        }
    }

    public double getWallet() {
        return wallet;
    }
```

```
    public void addMoneyToWallet(double amount) {
        wallet += amount;
    }

    public void deductMoneyFromWallet(double amount) {
        if (wallet < amount) {
            System.out.println("Insufficient balance in your wallet.");
            return;
        }
        wallet -= amount;
        System.out.println("Amount of INR " + amount + " deducted from your
wallet.");
    }

    public void rentCamera(Camera camera, int days) {
        if (wallet < camera.getPrice() * days) {
            System.out.println("Insufficient balance in your wallet.");
            return;
        }
        camera.setStatus("RENTED");
        double rentAmount = camera.getPrice() * days;
        wallet -= rentAmount;
        System.out.println("Amount of INR " + rentAmount + " deducted from your
wallet for renting camera with ID " + camera.getId() + ".");
```

```
    }
  }
```